Discontinuity of the weighted harmonic Hardy constant

A counterexample to MathDB problem 345844

Principia Math in Progress

September 12, 2026

Task 1. Resolve the continuity conjecture. Conjecture 1 of Pasteczka [1, p. 1061] asserts that a symmetric, monotone weighted mean, continuous in its weights and having the Hardy property, has a Hardy constant continuous for pointwise convergence of positive weight sequences. We refute this with the weighted harmonic mean on $I = (0, \infty)$.

For positive weights $w = (w_n)_{n\ge 1}$, put $W_n = \sum_{i=1}^{n} w_i$ and

$$M_n(x; w) = \frac{W_n}{\sum_{i=1}^{n} w_i/x_i}.$$

Write $\mathscr{H}(w)$ for the least $C \in [0, \infty]$ such that

$$\sum_{n\ge 1} w_n M_n(x; w) \le C \sum_{n\ge 1} w_n x_n \quad \text{whenever } x_n > 0, \quad \sum_{n\ge 1} w_n x_n < \infty. \tag{1}$$

This is the source's Definition 2.3. The harmonic mean is symmetric under simultaneous permutation of inputs and weights, monotone in each input, and continuous in the nonnegative weights when their sum is positive. Its formula also gives weight homogeneity, reduction, zero-weight elimination, and the minimum–maximum bounds required of a weighted mean.

Lemma 1. *For every positive weight sequence w, one has $\mathscr{H}(w) \le 9/2$.*

Proof. The Cauchy–Schwarz inequality and a right-endpoint integral bound give

$$\left(\sum_{i=1}^{n} w_i \sqrt{W_i}\right)^2 \le \left(\sum_{i=1}^{n} \frac{w_i}{x_i}\right)\left(\sum_{i=1}^{n} w_i W_i x_i\right), \qquad \sum_{i=1}^{n} w_i \sqrt{W_i} \ge \int_0^{W_n} \sqrt{t}\, dt = \frac{2}{3} W_n^{3/2}.$$

Consequently,

$$M_n(x; w) \le \frac{9}{4W_n^2} \sum_{i=1}^{n} w_i W_i x_i. \tag{2}$$

Since t^{-2} decreases, for every $i \ge 1$,

$$\sum_{n=i}^{\infty} \frac{w_n}{W_n^2} \le \frac{w_i}{W_i^2} + \sum_{n=i+1}^{\infty} \int_{W_{n-1}}^{W_n} \frac{dt}{t^2} \le \frac{2}{W_i}. \tag{3}$$

Tonelli's theorem, (2), and (3) show that the left side of (1) is at most

$$\frac{9}{4} \sum_{i\ge 1} w_i W_i x_i \sum_{n\ge i} \frac{w_n}{W_n^2} \le \frac{9}{2} \sum_{i\ge 1} w_i x_i.$$

In particular the harmonic mean satisfies the conjecture's Hardy hypothesis, even in the formulation requiring every positive weight sequence to have a finite Hardy constant. □

Lemma 2. *If $0 < w_n \le B < \infty$ and $W_n \to \infty$, then $\mathscr{H}(w) \ge 2$.*

Proof. Put $Q_n = \sum_{i=1}^{n} w_i^2$. The identity

$$\sum_{i=1}^{n} w_i W_i = \frac{W_n^2 + Q_n}{2} \tag{4}$$

follows by separating diagonal and off-diagonal terms in W_n^2. Fix $L \ge 1$ and $\delta > 0$, and use the strictly positive input

$$x_i = \frac{1}{W_i} \ (i \le L), \qquad x_i = \frac{\delta\, 2^{-(i-L)}}{w_i} \ (i > L).$$

Its weighted sum is $D_L + \delta$, where $D_L = \sum_{n=1}^{L} w_n/W_n$. By (4), its first L output terms have weighted sum

$$T_L = \sum_{n=1}^{L} \frac{2w_n W_n}{W_n^2 + Q_n}.$$

Thus $\mathscr{H}(w) \ge T_L/(D_L + \delta)$, and letting $\delta \downarrow 0$ gives $\mathscr{H}(w) \ge T_L/D_L$.

For $t \in [W_{n-1}, W_n]$ we have $t + B \ge W_n$. Hence

$$D_L \ge \int_0^{W_L} \frac{dt}{t+B} = \log(1 + W_L/B) \longrightarrow \infty.$$

On the other hand $Q_n \le BW_n$, so (3) with $i = 1$ yields

$$0 \le 2D_L - T_L = \sum_{n=1}^{L} \frac{2w_n Q_n}{W_n(W_n^2 + Q_n)} \le 2B \sum_{n=1}^{L} \frac{w_n}{W_n^2} \le \frac{4B}{w_1}.$$

It follows that $T_L/D_L \to 2$, proving the assertion. □

Theorem 3. *The conjectured continuity fails, even among strictly positive integer weight sequences with infinite total mass.*

Proof. For $n, K \ge 1$, define

$$\lambda_n = 4^{\,n-1}, \qquad \mu_n^{(K)} = \begin{cases} 4^{\,n-1}, & n \le K, \\ 4^{\,K-1}, & n > K. \end{cases}$$

Both sequences have infinite total mass, and $\mu^{(K)} \to \lambda$ pointwise: each fixed coordinate agrees exactly once $K \ge n$. Each $\mu^{(K)}$ is bounded, so Lemma 2 gives $\mathscr{H}(\mu^{(K)}) \ge 2$. For the limiting sequence,

$$M_n(x;\lambda) \le \frac{\Lambda_n}{\lambda_n} x_n = \frac{4 - 4^{\,1-n}}{3} x_n \le \frac{4}{3} x_n.$$

Multiplication by λ_n and summation give $\mathscr{H}(\lambda) \le 4/3$. Thus

$$\mathscr{H}(\mu^{(K)}) - \mathscr{H}(\lambda) \ge \frac{2}{3} \quad \text{for every } K,$$

which contradicts continuity. Lemma 1 and the stated properties of the harmonic mean verify every hypothesis. □

Sharper values and verification. The proof above is self-contained and needs no sharp Hardy theorem. As a consistency check, the source's Lemma 4.1, with Gini parameters $(p, q) = (-1, 0)$, gives $\mathscr{H}(\lambda) = 5/4$ and $\mathscr{H}(\mu^{(K)}) = 2$: the respective nonincreasing ratios w_n/W_n tend to $3/4$ and 0. The accompanying exact rational checker verifies finite positive-input witnesses, identity (4), and the stated bounds. Finite tests are supplementary; the proof covers every K and infinite sequences.

Task 2. Classify weights attaining the maximal constant. Find explicit necessary and sufficient conditions on a positive weight sequence for its harmonic Hardy constant to equal 2, including irregular sequences outside the monotone-ratio setting.

Task 3. Determine constants for oscillating weight growth. Compute the harmonic Hardy constant when w_n/W_n has two or more accumulation points, and determine which features beyond those points affect the answer.

Task 4. Find useful topologies restoring continuity. Identify a topology stronger than coordinatewise convergence that permits meaningful infinite-tail perturbations and makes weighted Hardy constants continuous for broad classes of Hardy means.

Task 5. Quantify the effect of a tail replacement. For harmonic and other power means, bound the change in the Hardy constant under controlled perturbations of all weights after a specified index, with explicit dependence on the perturbation size.

References

[1] P. Pasteczka, *On properties of weighted Hardy constant for means*, Mathematical Inequalities & Applications **25**(4) (2022), 1047–1068. doi:10.7153/mia-2022-25-65; published text; arXiv:2003.06025. Definitions: pp. 1051–1052. Conjecture 1 and Lemma 4.1: p. 1061.